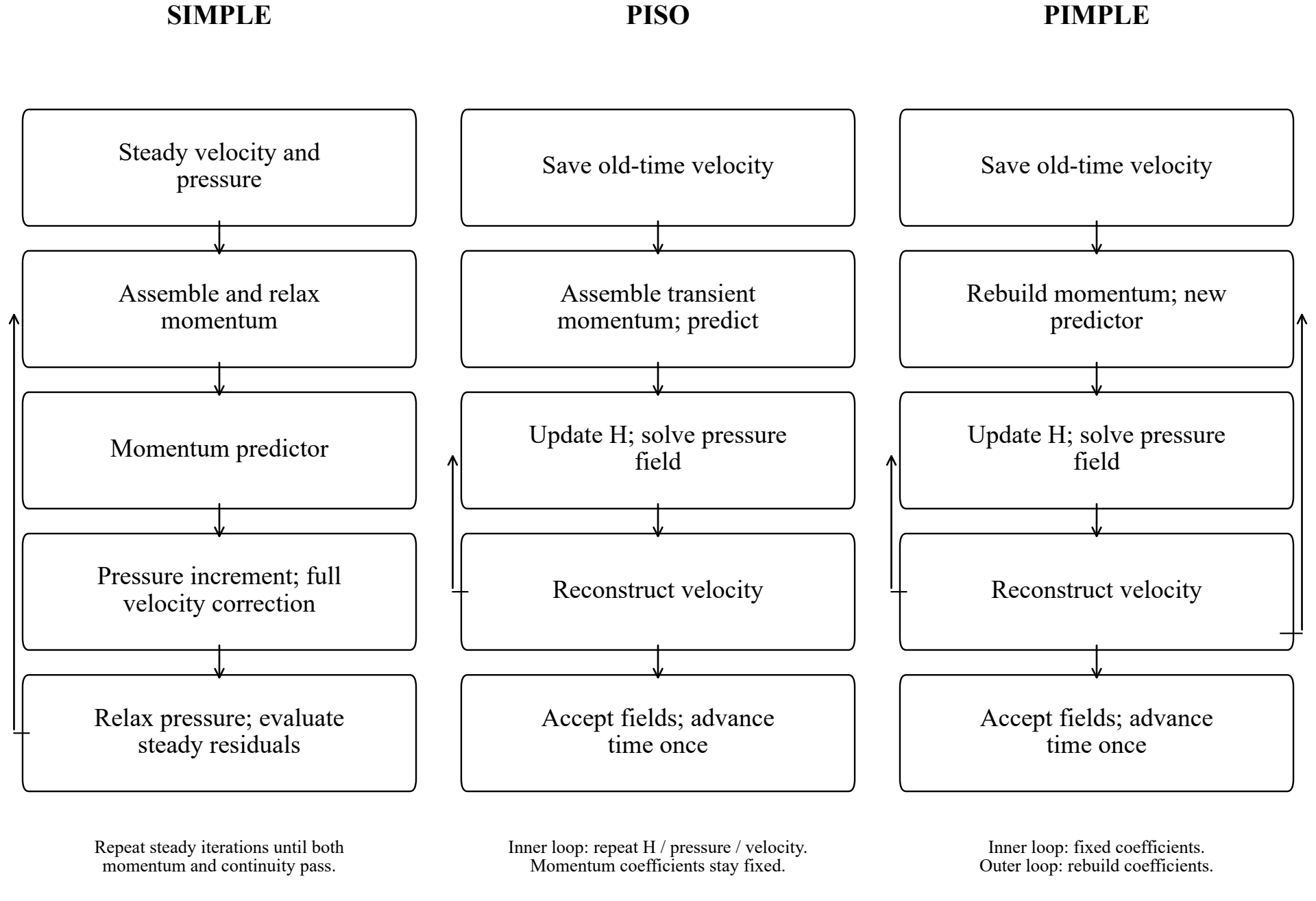
SIMPLE
Steady velocity and pressure
Assemble and relax momentum
Momentum predictor
Pressure increment; full velocity correction
Relax pressure; evaluate steady residuals
Repeat steady iterations until both momentum and continuity pass.
PISO
Save old-time velocity
Assemble transient momentum; predict
Update H; solve pressure field
Reconstruct velocity
Accept fields; advance time once
Inner loop: repeat H / pressure / velocity.
Momentum coefficients stay fixed.
PIMPLE
Save old-time velocity
Rebuild momentum; new predictor
Update H; solve pressure field
Reconstruct velocity
Accept fields; advance time once
Inner loop: fixed coefficients.
Outer loop: rebuild coefficients.